

Micro-Manager workshop

- If you haven't installed **Micro-Manager** and **uv** yet:
- **Windows, macOS:** follow instructions at <https://github.com/uw-loci/mm-workshop-i2k-bina-2026>
(Link also in Swapcard)
 - **For macOS, a quick extra step was added yesterday to patch a bug**
- **Linux:** it's probably too late (build & install is complicated)
- You should be able to follow along even if you cannot try it out yourself (slides are available at the same link)



Micro-Manager from GUI to Python: Interactive Control, Configuration, and Scripted Data-Driven Acquisition

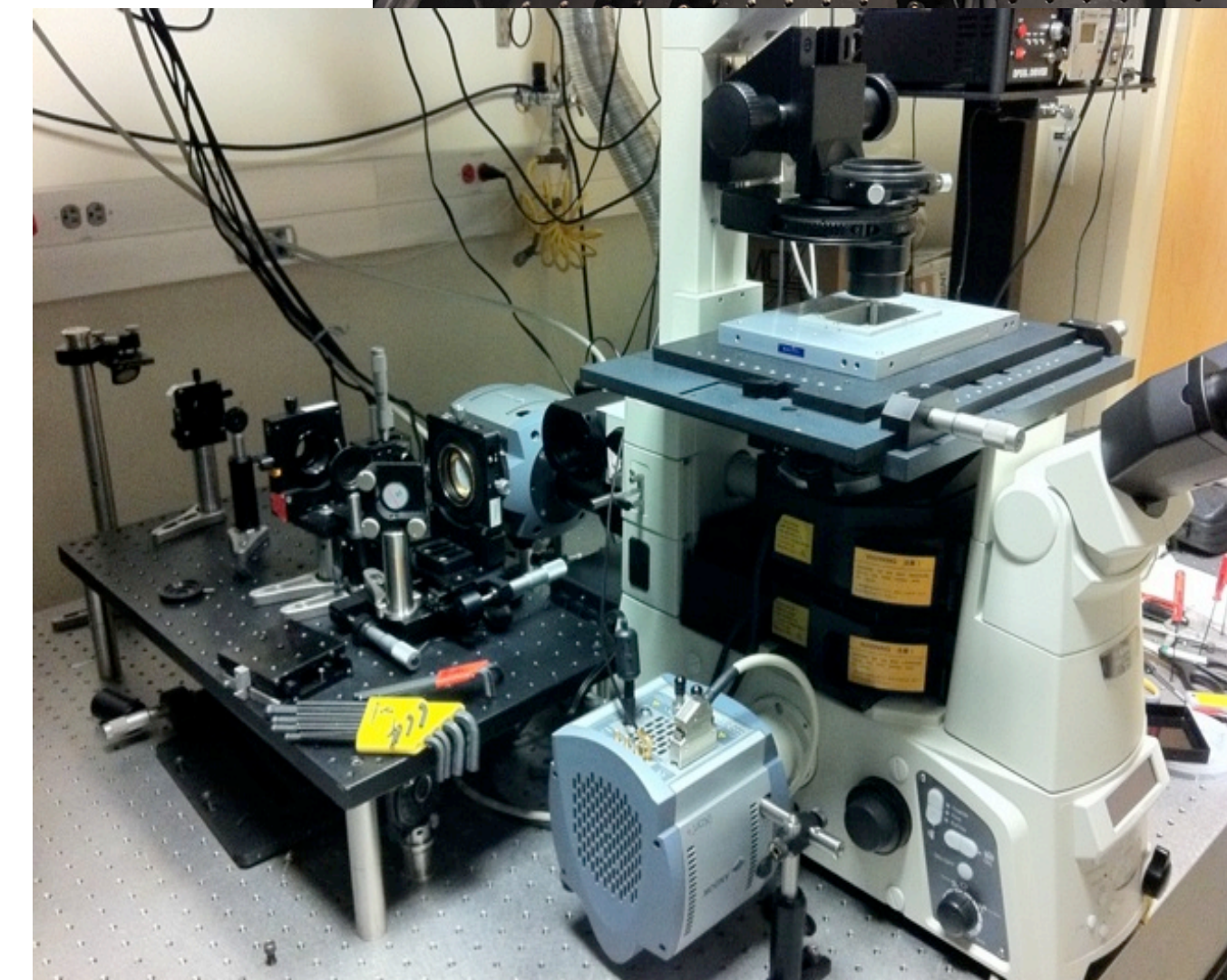
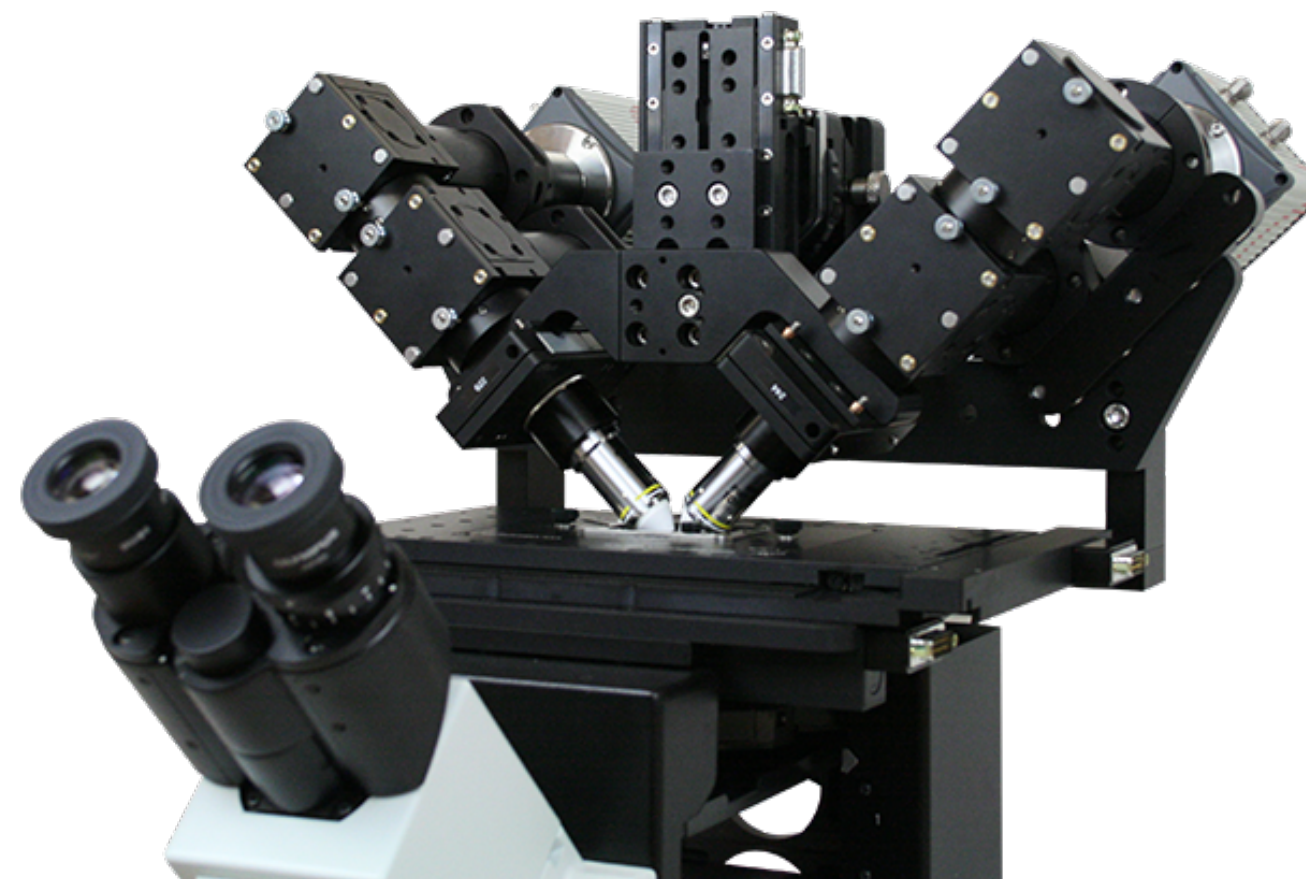
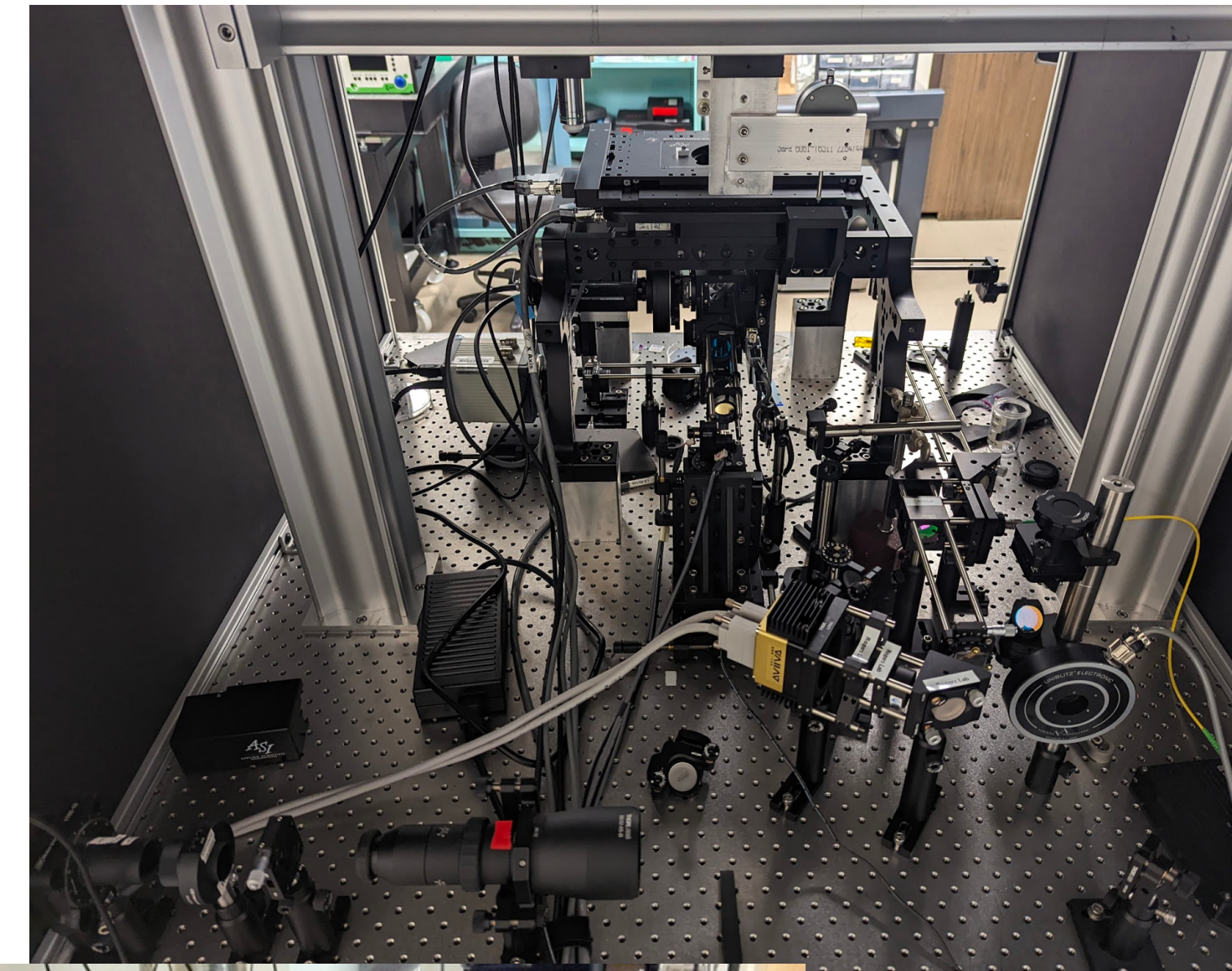
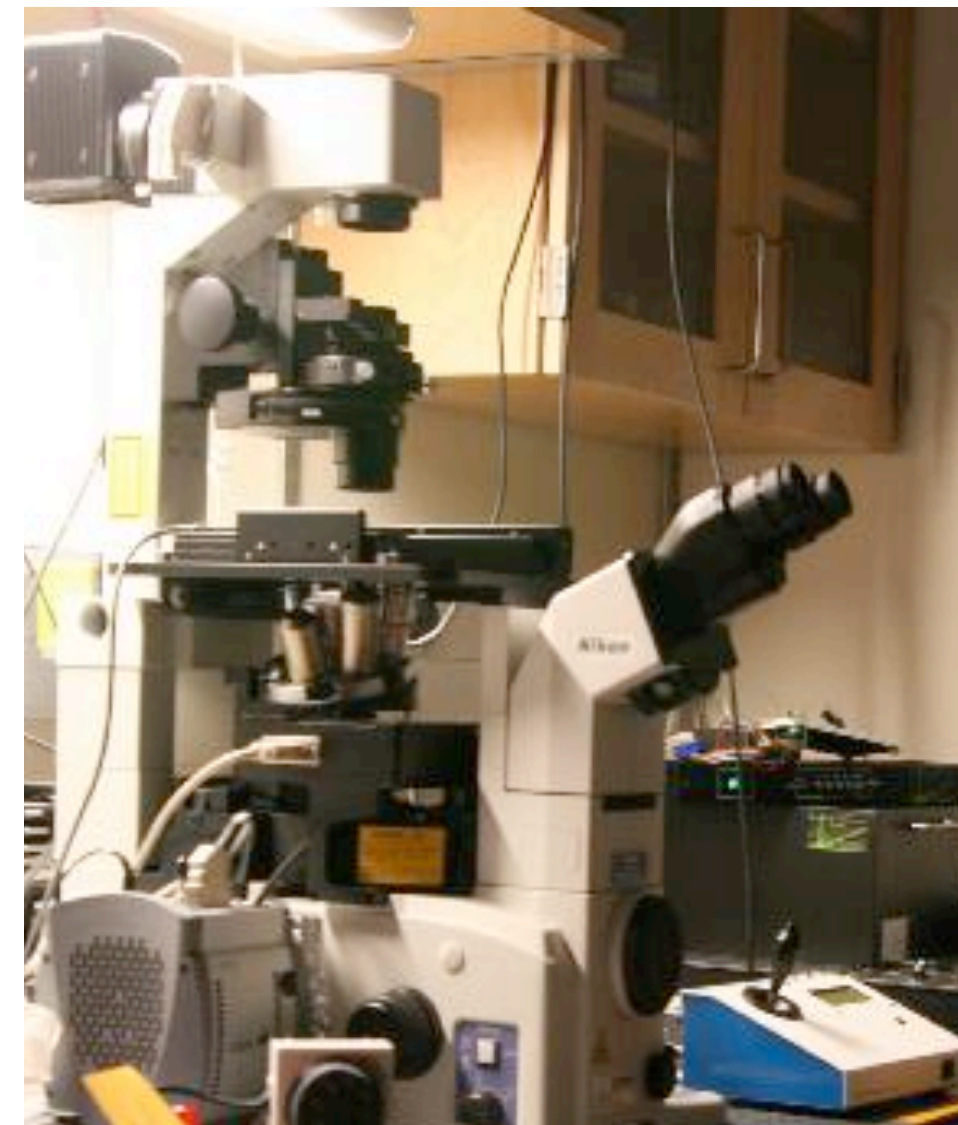
Mark A. Tsuchida and Gabriel Selzer
University of Wisconsin–Madison

2026.09.29

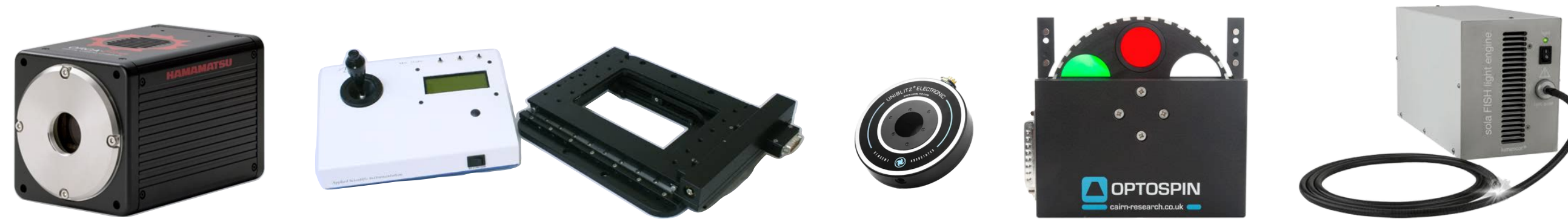
Workshop overview

- Part 1 (Mark)
 - Basics of computer control of microscopes
 - Micro-Manager basics, configuration, multi-dimensional acquisition
- Part 2 (Gabriel; intro by Mark)
 - Python scripting with pymmcore-plus
 - Data-driven acquisition demo

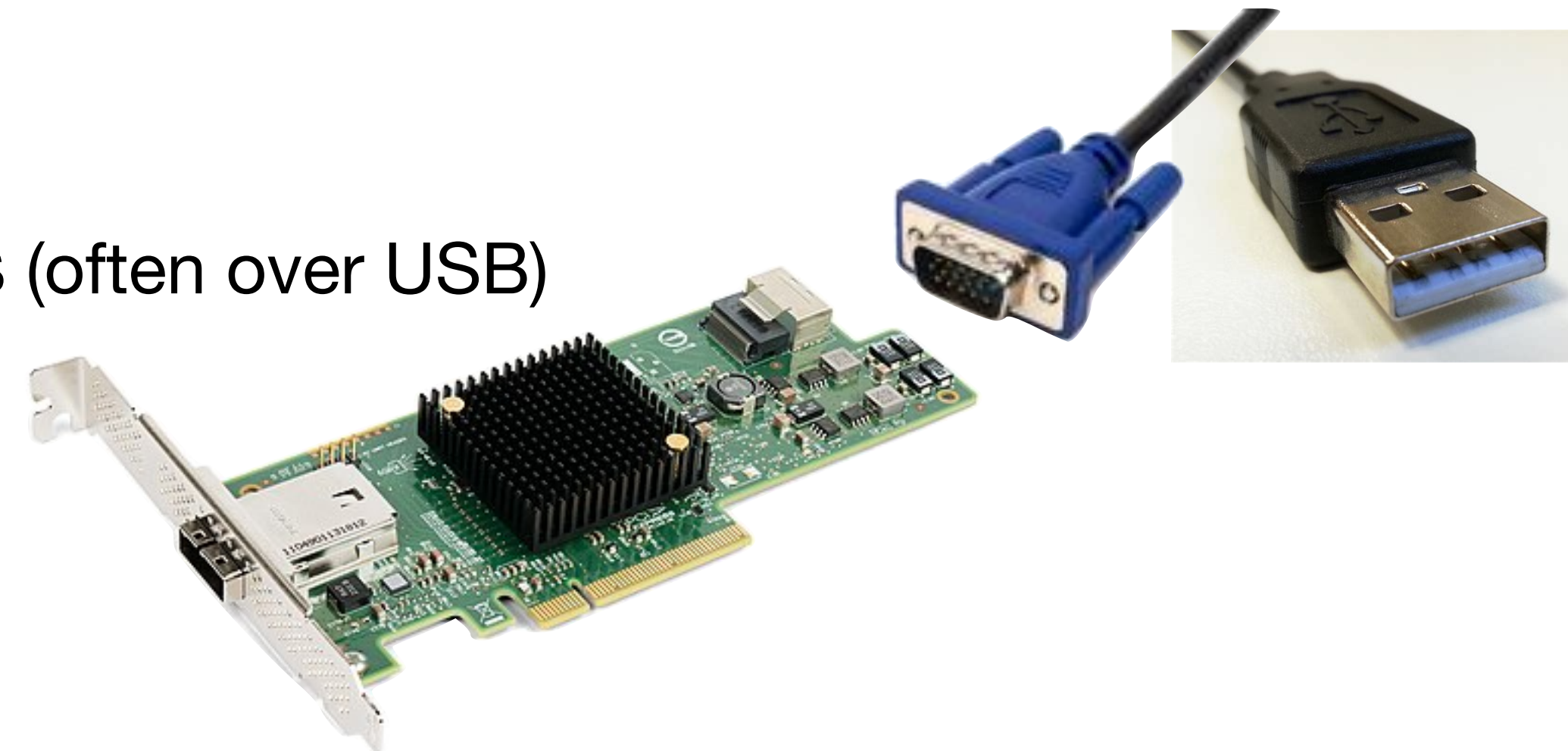
Motorized/automated research microscopes



Computer control: what the PC (software) sees

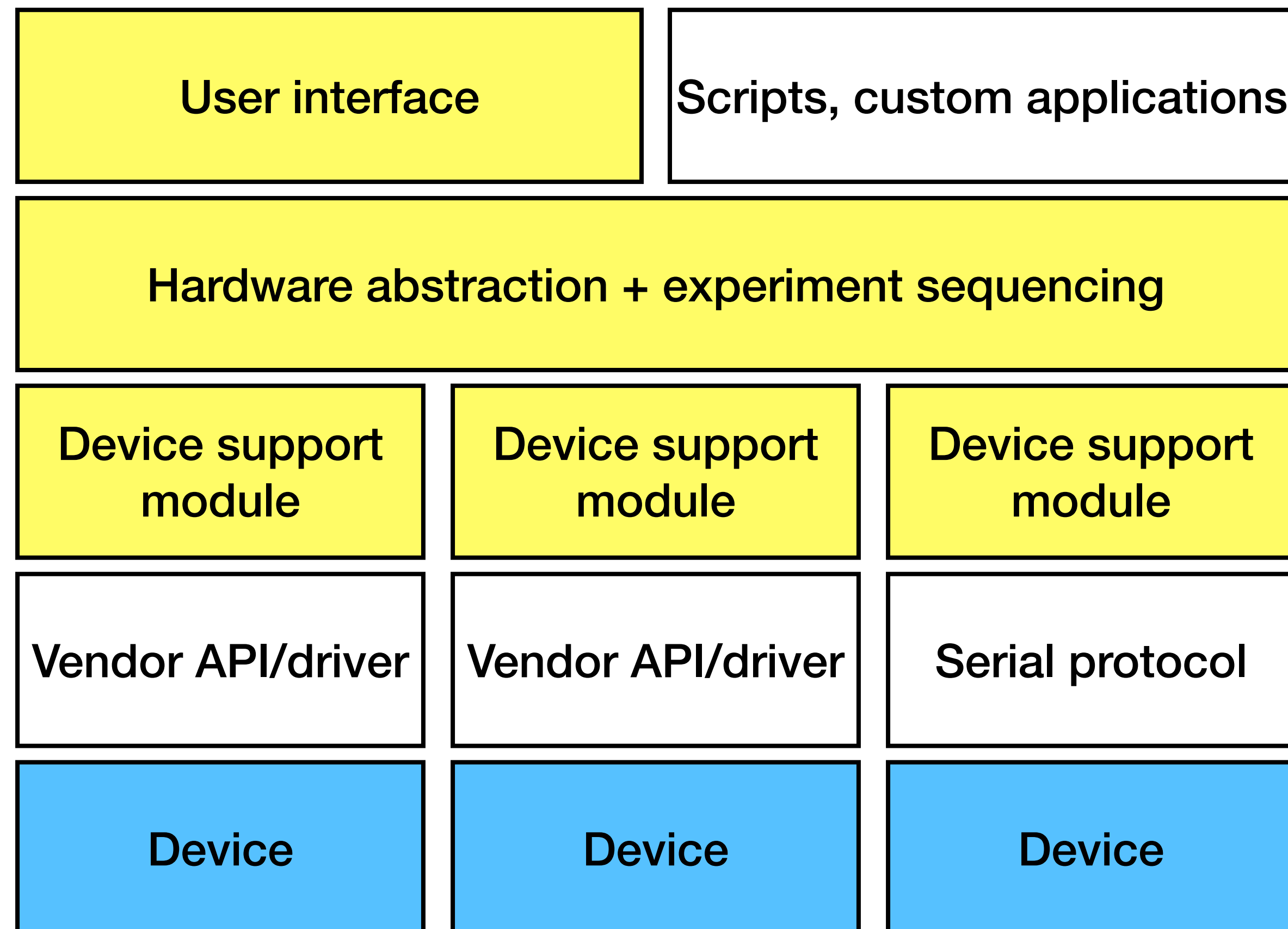


- Devices: cameras, stages, shutters, wheels, light sources, control units, SLMs, focus maintenance, ...
 - **Integrated** into microscope stand/control box vs **stand-alone** devices
- Connection / software interface
 - **Serial** port (COM port, RS-232 port): stream of bytes (often over USB)
 - PCIe, USB + **driver** (usually proprietary) + **API**
- **No standard interfaces***, unlike consumer products



Software stack for microscope control

A general mental model



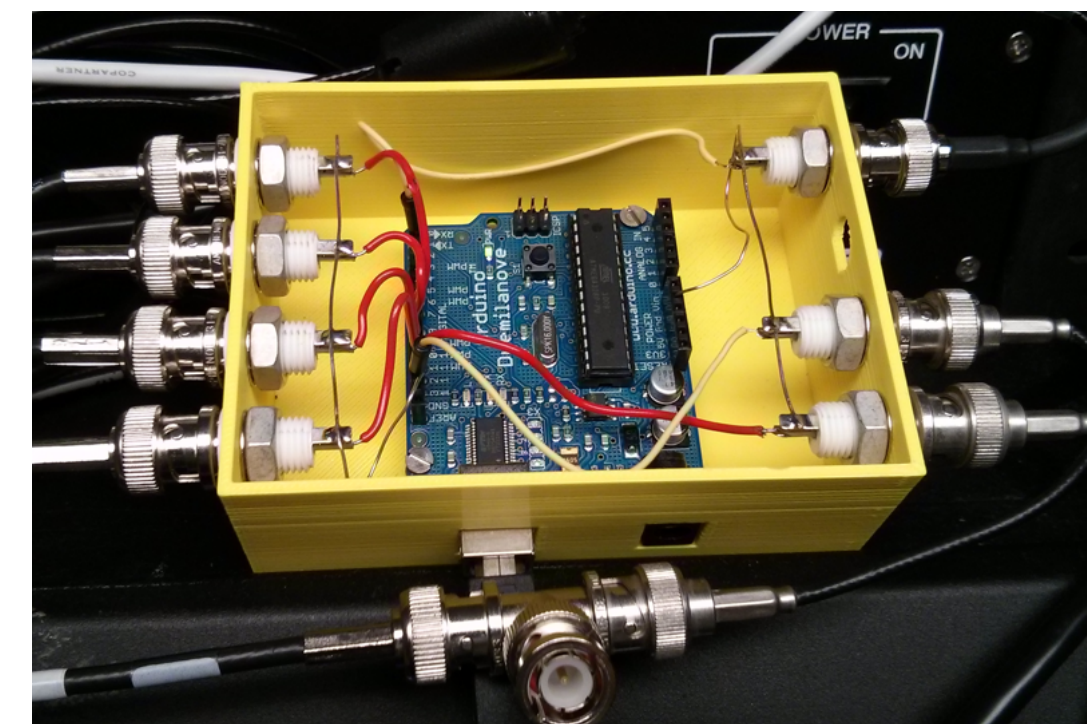
Acq Software
Hardware

User says: “40x, DAPI, 100 ms”

- Put devices into position:
 - **Objective turret** position 3 (40x 0.75 Air)
 - **Filter cube turret** position 5 (DAPI/FITC/TRITC)
 - Select **LED line** 1 (405 nm)
 - **Camera** exposure = 100 ms
 - Active **shutter** = epi path shutter (or the LED itself)
- **Open shutter** → (wait) → **snap image** → (wait) → **close shutter**
- Read out and return the **image**

Speaking of waiting

- **Software-timed/sequenced**
 - Control returns to software at every synchronization point
 - Subject to communication/OS overhead & jitter (with long tail)
- **Hardware-timed/sequenced**
 - Direct timing signals (triggers) between hardware devices
 - Deterministic timing; low overhead — but less flexible



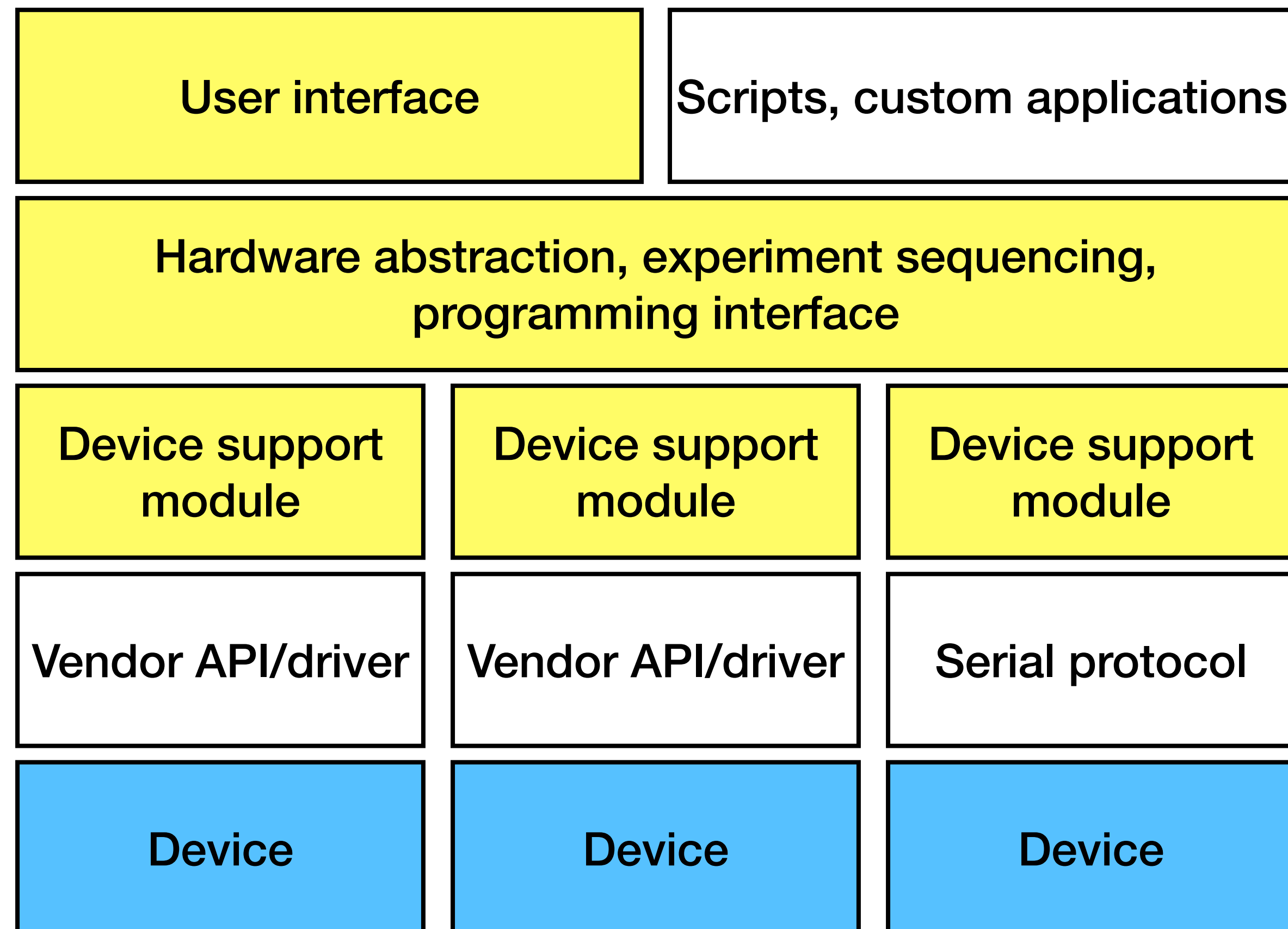


- Microscope control & acquisition software
- Since 2005: Nico Stuurman & Nenad Amodaj (Ron Vale lab, then at UCSF)
- Open source, vendor-neutral, open-ended
 - Same basic user interface for any microscope
 - Customize, extend, and share (building blocks for special techniques)
 - Equipment manufacturers (and scientists) can add support for their device, enabling integration with a full microscope system

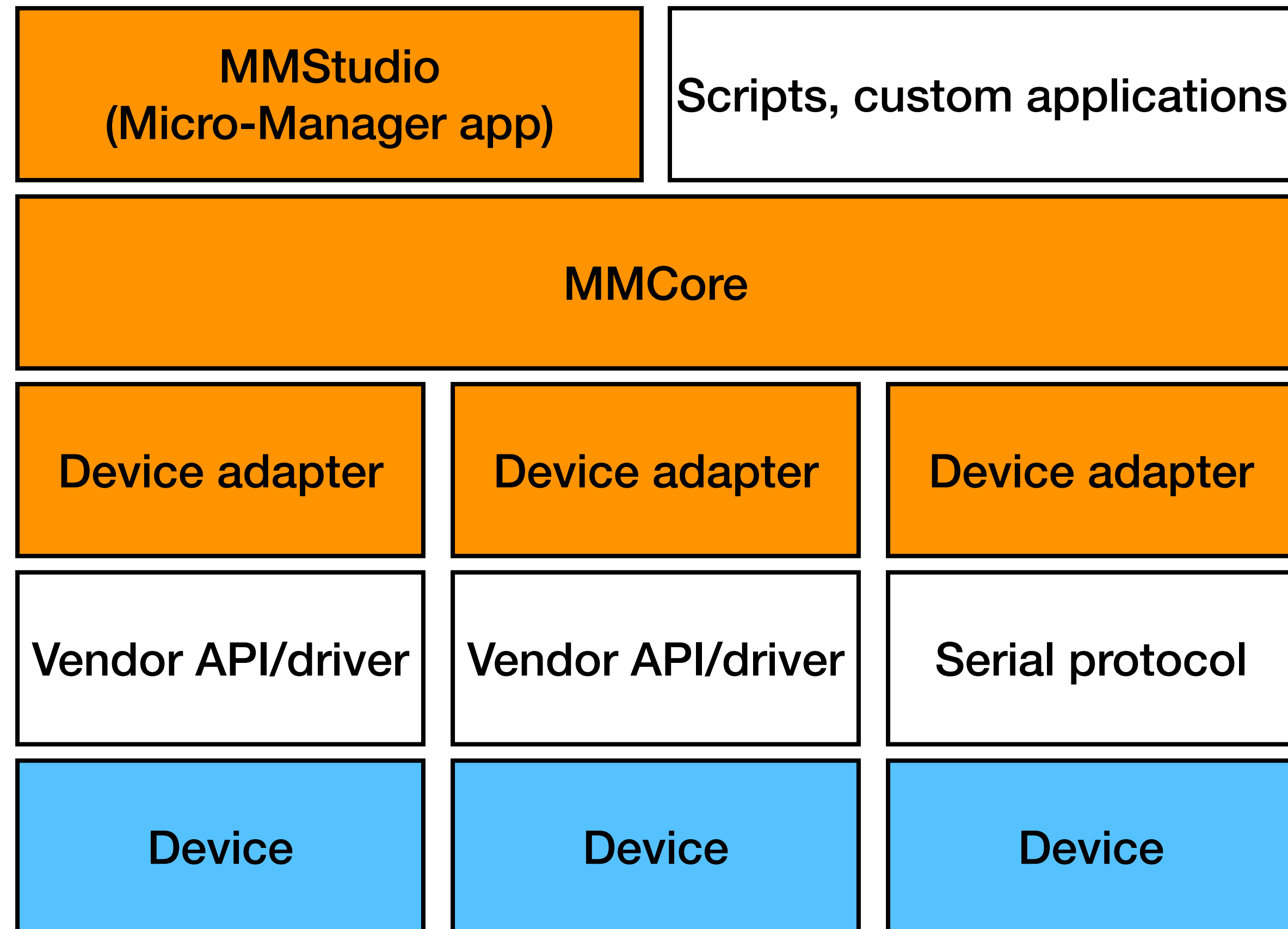


Nico Stuurman Nenad Amodaj
& many contributors over the years

Software stack for microscope control



Micro-Manager architecture



First look

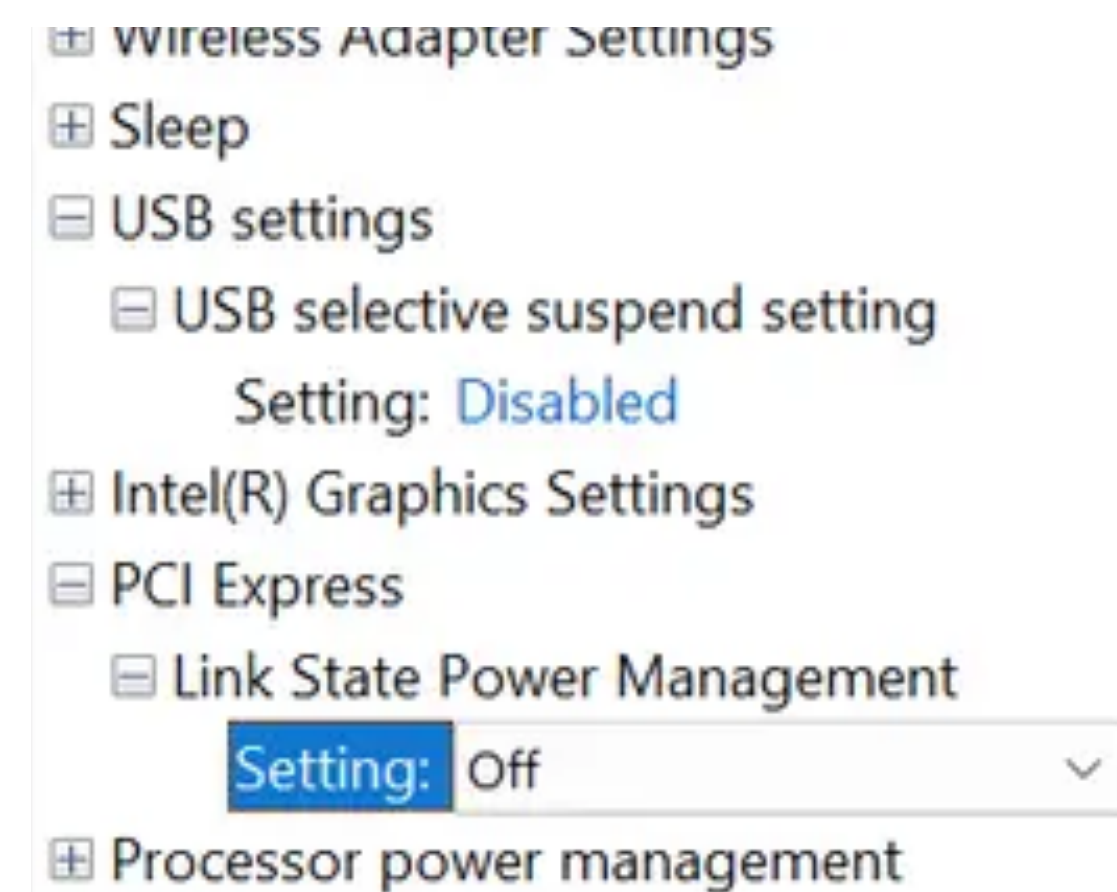
- Demo

First look: hands-on

- Launch Micro-Manager
- Select hardware configuration: **Simulated-full.cfg**
<https://github.com/uw-loci/mm-workshop-i2k-bina-2026/blob/main/Simulated-full.cfg>
- Try: **Snap, Live**, change **Exposure**, adjust **histograms**
- Try: **Autoshutter**
- Try: **Stage Control**

Installing & configuring MM (for a real microscope)

- **Windows** is common/easiest/often required (wide use + driver availability)
- **Nightly build** from <https://download.micro-manager.org/>
- Some devices need **vendor drivers**: check device pages https://micro-manager.org/Device_Support (just install / also copy DLL)
- Test device with vendor app first, if available
- Good idea to disable Windows **USB selective suspend setting**, **PCI Express link state power management**
(Control Panel > System and Security > Power Options > Edit Plan Settings > Change advanced power settings)



Hardware Configuration Wizard

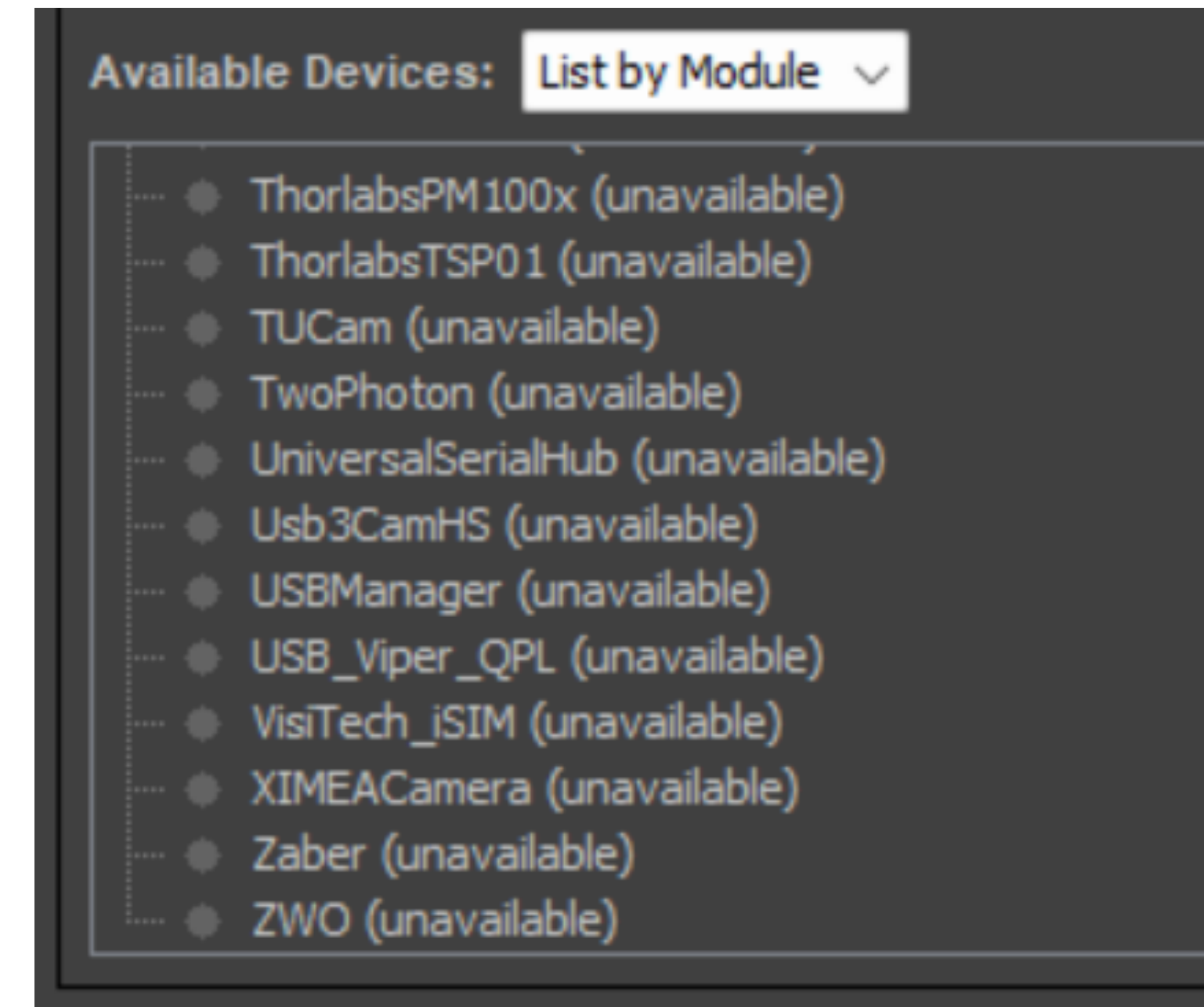
- Demo: configure SimulatedMicroscope

Hardware config: hands-on

- **Devices > Hardware Configuration Wizard...**
Create new configuration
- Add **SimulatedMicroscope > SimHub**
Select all peripherals: SimCam, SimFocus, SimObjectiveTurret, SimXY, SimShutter
- Default devices: the defaults are fine
- Delays: not applicable to SimulatedMicroscope
- Position labels: accept the defaults
- Name and save (e.g.: MySimulated.cfg)
- Confirm that Snap works

Hardware config pitfalls

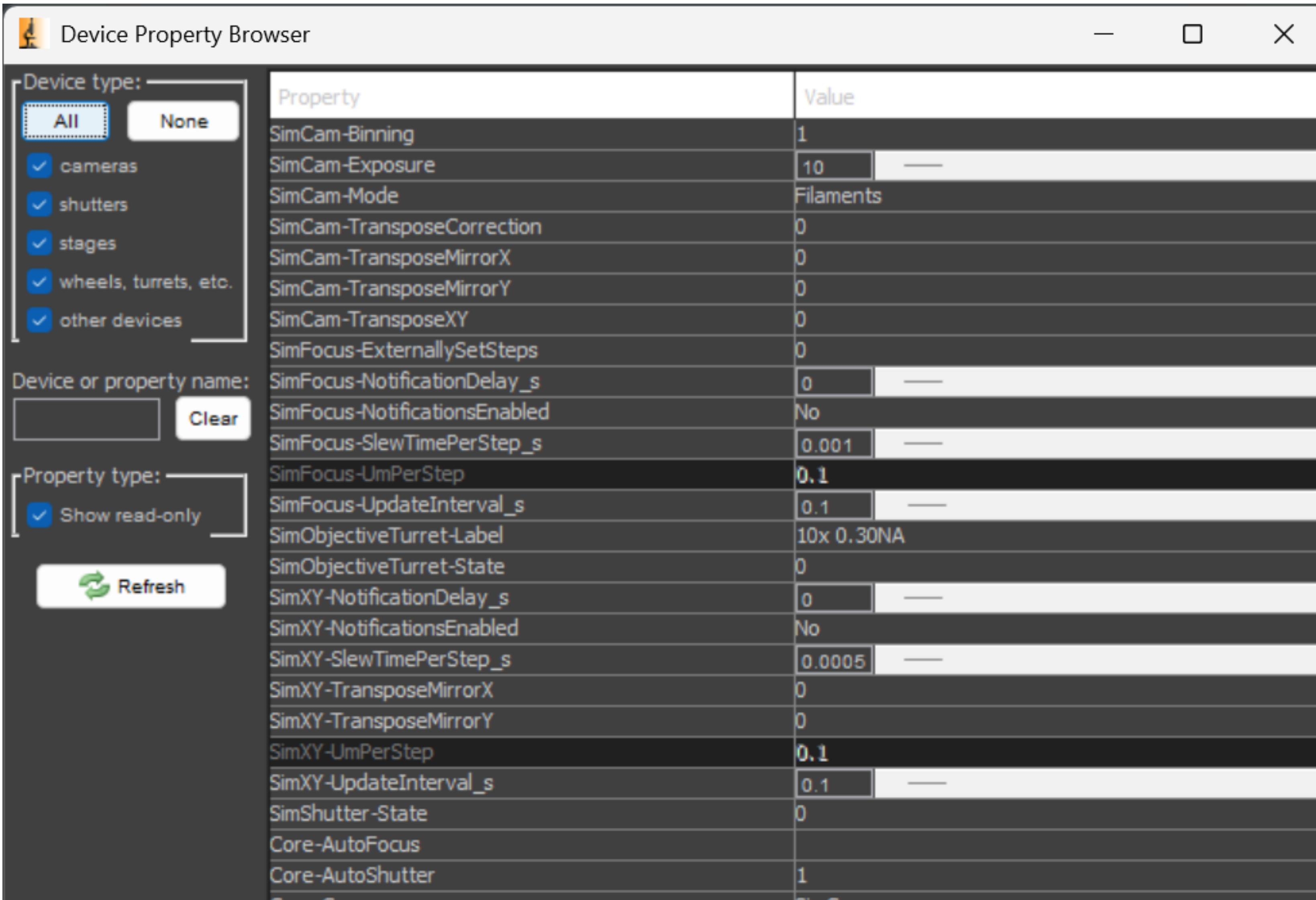
- Vendor/third-party **driver** not installed
 - Or not in correct **location**, or wrong **version**
- Separately obtained **device adapter** DLL:
Device Interface Version mismatch
- **Serial port** settings (check device page, equipment manual, or ask)
- **Other software** (e.g., vendor app) has control (quit before running MM)



Troubleshooting starting points

- [https://micro-manager.org/Device Support](https://micro-manager.org/Device_Support)
- CoreLog and **Help > Create Problem Report...** (quick demo)
- <https://forum.image.sc/> (most user questions; use tag **micro-manager**)

Device settings are properties



The screenshot shows a window titled "Device Property Browser". On the left, there are filters for "Device type" (All, None, cameras, shutters, stages, wheels, turrets, etc., other devices) and "Property type" (Show read-only). A "Refresh" button is at the bottom left. The main area is a table with two columns: "Property" and "Value".

Property	Value
SimCam-Binning	1
SimCam-Exposure	10
SimCam-Mode	Filaments
SimCam-TransposeCorrection	0
SimCam-TransposeMirrorX	0
SimCam-TransposeMirrorY	0
SimCam-TransposeXY	0
SimFocus-ExternallySetSteps	0
SimFocus-NotificationDelay_s	0
SimFocus-NotificationsEnabled	No
SimFocus-SlewTimePerStep_s	0.001
SimFocus-UmPerStep	0.1
SimFocus-UpdateInterval_s	0.1
SimObjectiveTurret-Label	10x 0.30NA
SimObjectiveTurret-State	0
SimXY-NotificationDelay_s	0
SimXY-NotificationsEnabled	No
SimXY-SlewTimePerStep_s	0.0005
SimXY-TransposeMirrorX	0
SimXY-TransposeMirrorY	0
SimXY-UmPerStep	0.1
SimXY-UpdateInterval_s	0.1
SimShutter-State	0
Core-AutoFocus	
Core-AutoShutter	1

Not *everything* is a property:
XY/Z position, camera
acquisition, ...

Configuration groups & presets

- Often, we need to move/set several device positions/parameters together
 - Recall 40x DAPI example: turrets, illumination line, active shutter
- **Configuration preset** = a set of properties and chosen values
- **Configuration group** = a collection of presets over the same properties
- Use for: channels, light path, objective/magnification, ...
- The **channel group** is assigned a special meaning

Config groups/presets

- Demo
 - **Channel** group (incl. Core-Shutter property = active shutter)
 - **Objective** group (single-property special case)

Config groups/presets: hands-on

- (Starting point: Simulated-hw.cfg)
- **Group:** + → name "**Channel**"; select **SimCam-Mode** and **Core-Shutter**
 - Preset name "**Rhodamine**"; Mode = **Filaments**; Shutter = **SimShutter**
- Select "**Channel**" group; **Preset:** + → name "**DAPI**"; Mode = **Nuclei**
- **Save**

Config groups/presets: pitfalls/tips

- **Don't include shutter state** in the channel group
 - Shutter is specially handled by Autos shutter
- **Don't include camera exposure** in the channel group
 - There is a built-in mechanism to remember per-channel exposure
- Try to keep groups **orthogonal** to each other where possible

Config groups/presets special cases

- Config group created with a **single property**
 - automatically display single-property control (slider if numeric)
- **Chanel group**: select in main window; used in a few slides
- Special config group "**System**"; presets "**Startup**" and "**Shutdown**"
 - apply after loading hardware config / before unloading

Pixel size calibration

- Demo (time permitting)
- Final config file with pixel size calibrations: `Simulated-full.cfg`

Pixel size calibration tips

- Automatic (stage-based) measurement is an optional convenience
 - You can type in the $\mu\text{m}/\text{px}$
- Do **not** include **Magnifier devices** (tube lens, optovar)
 - They are handled automatically
- Camera binning is also handled automatically



Multi-dimensional acquisition (MDA)

- The workhorse for most experiments
- Sequence time series, stage positions, Z stacks, channels (in different orders)

MDA

- Demo

MDA: hands-on

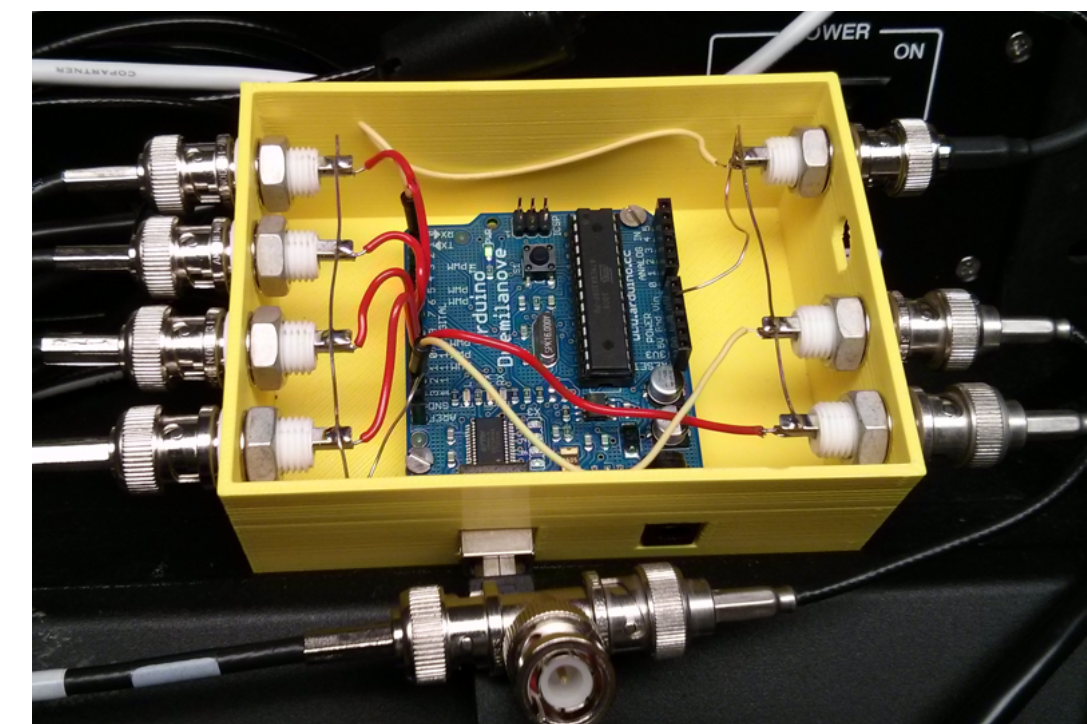
- Time + Channels
- Z-Stack + Channels

Opening saved image datasets

- Standard save format: "Image stack file"
(OME-TIFF + Micro-Manager metadata)
- Drag to Micro-Manager main window to re-open
- Drag the `.ome.tif` to Fiji (icon or toolbar) to open (Bio-Formats Importer)

Some additional highlights

- Plugins
 - Realtime Intensity Plot (a "typical" Micro-Manager plugin)
 - HCS Site Generator
 - Explorer (cutting edge!)
- Hardware triggered acquisitions



Software (normal)

4 channels, 10 time points



Hardware sequenced



Courtesy of Sam Lord

Z stack acquisitions displayed in real time



Courtesy of Sam Lord

Micro-Manager limitations

- Device support (even though we have 200+)
- Commercial laser-scanning microscopes not supported (spinning disk is fine)
- Commercial support/services may be harder to find
- "Very fast" acquisitions
 - High frame rate or data throughput (working on improvements)
 - Fast/deterministic feedback loops with PC in the loop (often need a dedicated design)

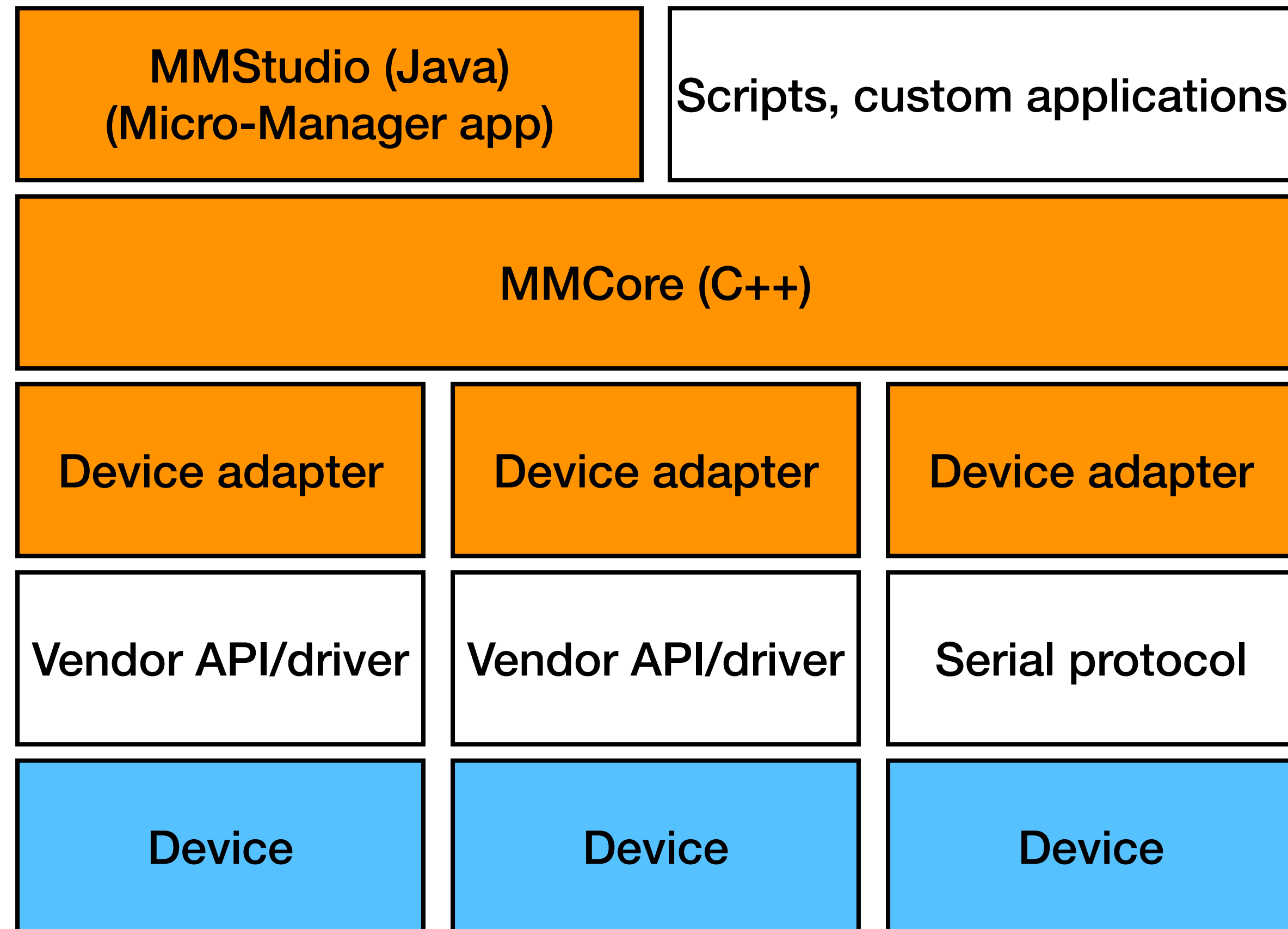
Getting help

- <https://micro-manager.org/> (you can edit it, too!)
- CoreLog and **Help > Create Problem Report...**
- <https://forum.image.sc/> (most user questions; use tag **micro-manager**)
- <https://github.com/micro-manager/micro-manager/>
(developer-centric: if you're sure it's a bug or missing feature)

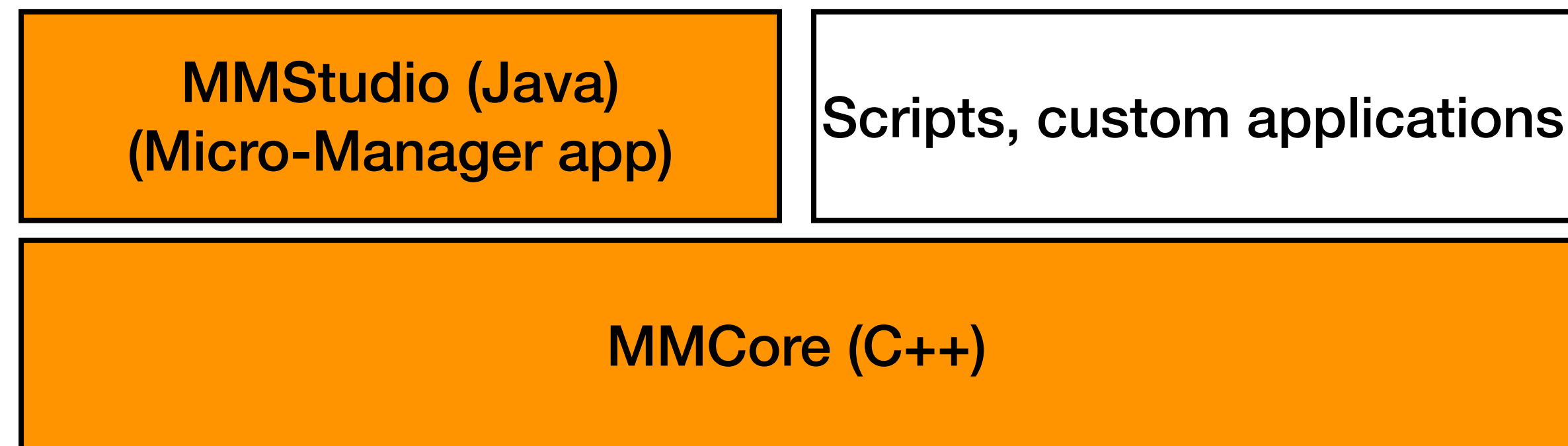
X-minute break

next: Python scripting

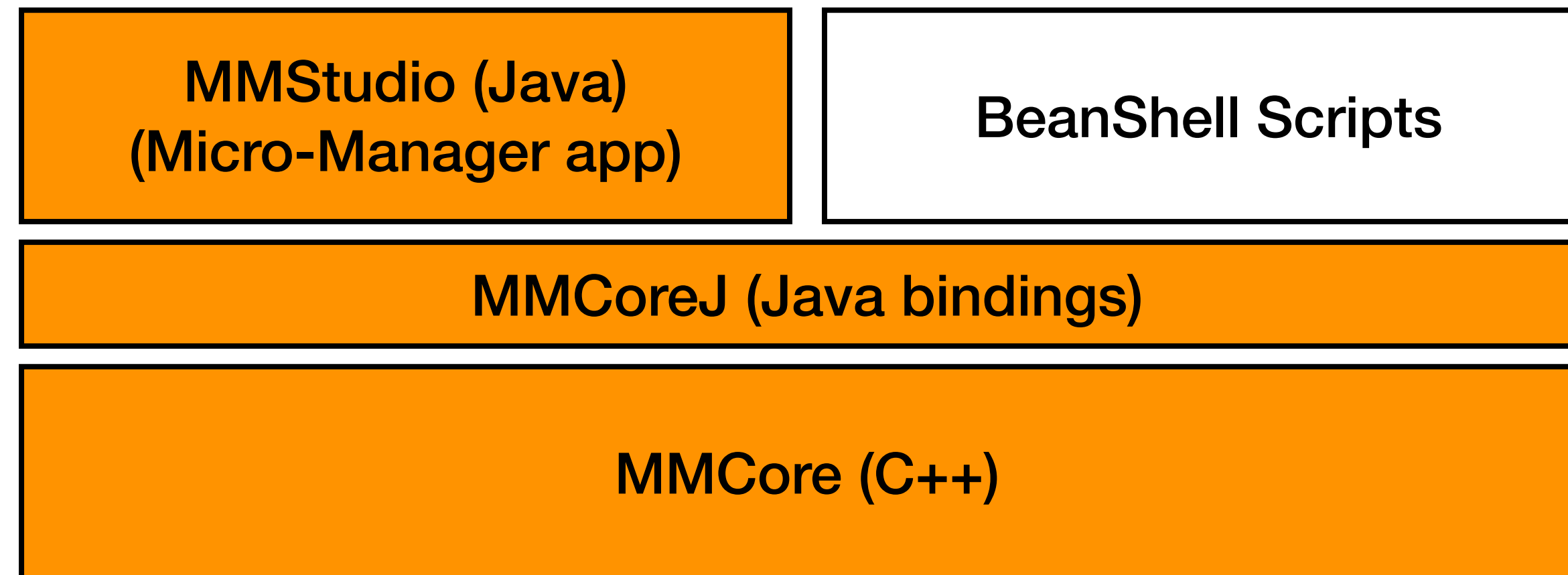
Micro-Manager: traditionally Java-based



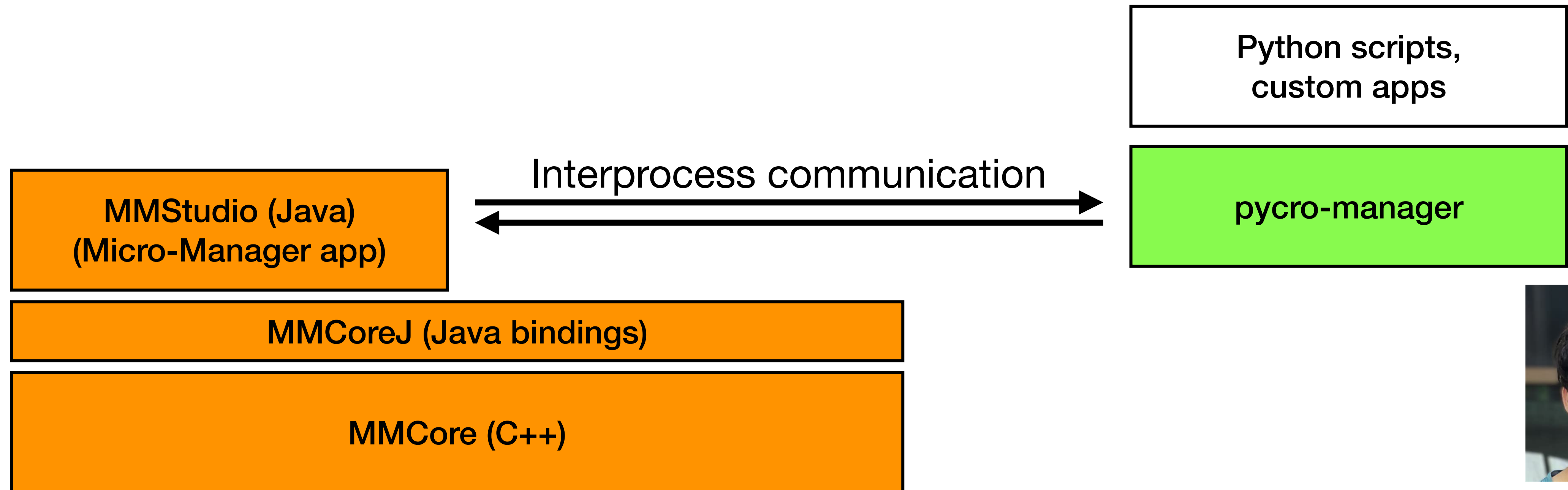
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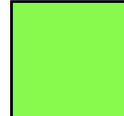


pycro-manager

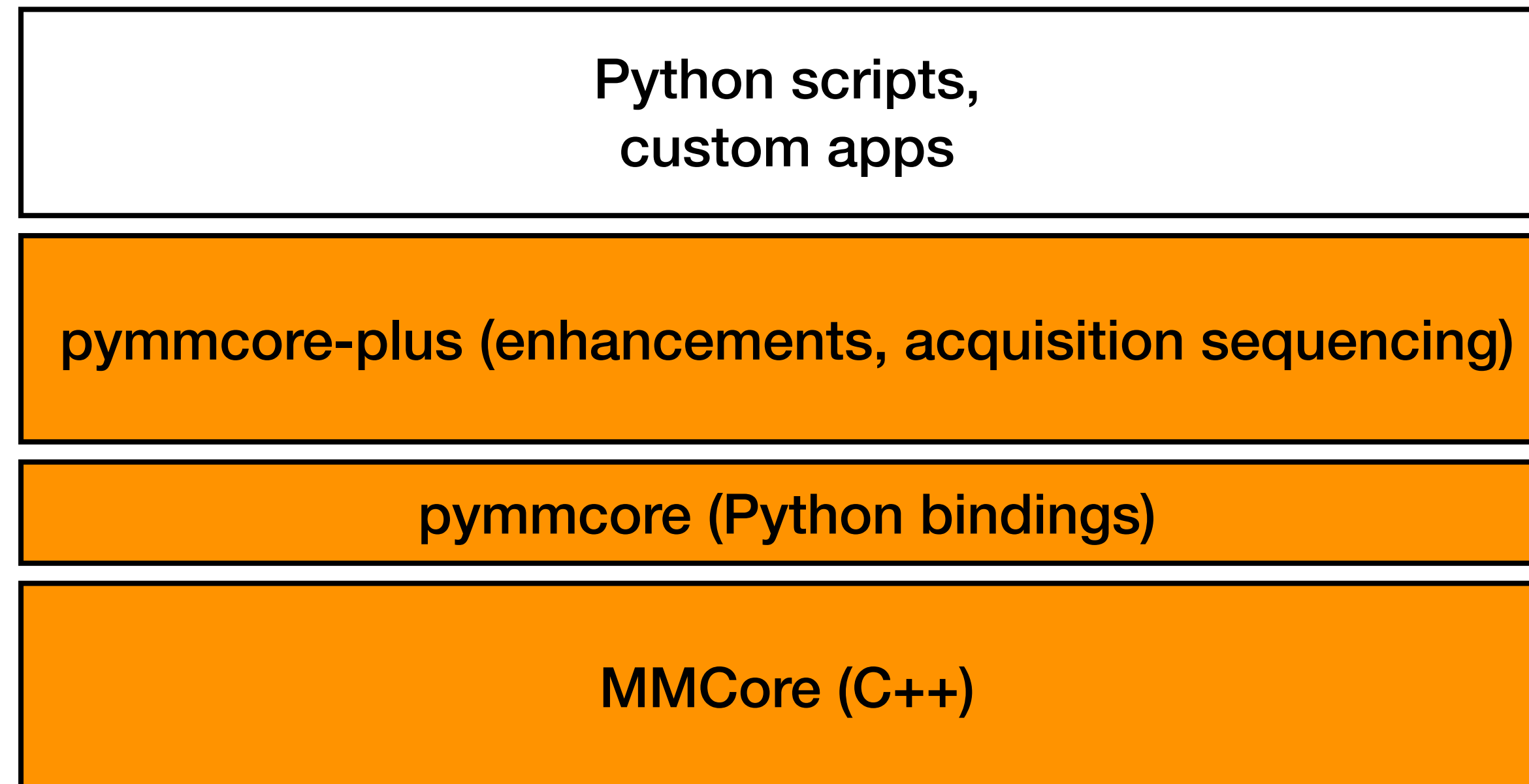


Henry
Pinkard

- Pragmatic architecture at the time; allows interactive use in the same session
- Indirection through Java means 3 languages involved
- Maintenance mode

 Python
 Micro-Manager

pymmcore-plus



- Direct architecture
- Philosophy: build up from reusable components (control, acquisition, UI)

pymmcore-plus ecosystem

- **pymmcore-plus**
- **useq-schema** — acquisition sequence as data
- **pymmcore-widgets** — reusable UI elements
- **napari-micro-manager**
- **ome-writers, ndv** (image viewer), ...
- **pymmcore-gui** (experimental full app)
- Device adapters in Python (experimental)



Talley
Lambert



Federico
Gasparoli



Gabriel
Selzer

<https://pymmcore-plus.github.io/>